



CODECHEF

The CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest, now in print!

*H*ello and welcome to the CodeChef Puzzle Challenge sponsored by LINUX For You and Direct! Let's see how many of you can solve this month's puzzle, and find out which one of you will be amongst the three lucky winners of a cash prize of Rs 1000!

The October edition's puzzle

Sometime back, Donald Knuth (if you don't know him, you need not read any further) was scheduled to make an "Earthshaking Announcement" at the TeX 32nd Anniversary Celebrations. It was widely speculated that he would announce the solution to the $P=NP$ problem, but that turned out to be a joke. Apu, a computer science student from India with no sense of humour, did not take this lightly and decided to solve the problem himself. Then he realised Vinay Deolikar had already solved the problem. So Apu decided to solve the "Sorting Problem" in less than $O(N \cdot \log(N))$ time, using something called a k -ordered array.

An array $A[1...n]$ is said to be k -ordered if $A[i - k] \leq A[i] \leq A[i + k]$ for all i such that $k \leq i < n - k + 1 \leq n$.

However, Apu is now stuck. What is the maximum possible distance between two numbers that are out of sorts, given a very long array—an array that is both 5-ordered and 7-ordered? (Distance implies the difference of indexes of the numbers, e.g., in this 2-ordered array [4, 1, 5, 2, 6, 3] the maximum distance is 5). Can you help him out?

The solution

The maximum possible distance between two

numbers that are out of sorts, given a very long array – an array which is both 5-ordered and 7-ordered—is 23.

How did we do it?

The given array is 5-sorted and 7-sorted. So, for any given number A at index I in the array, all the numbers at indices $(I + 5 \cdot X)$ ($X > 0$) will be greater than or equal to A —the same applies for all numbers at $(I + 7 \cdot X)$. Extending this a bit, we get that all numbers at $(I + 5 \cdot X + 7 \cdot Y)$ ($X, Y > 0$) will be in order. Hence, the desired answer is the largest number that cannot be expressed as $(5 \cdot X + 7 \cdot Y)$.

And the winners for October are...

No one :(

Your Codechef Puzzle Of The Month

A headmaster of a primary school performs an activity with the students of a class to encourage them to perform better in academics. He asks them to stand in queue, starts calling the students out one by one and asks them their rank in class.

Each one has a unique rank in class. If the rank of a student is better than his/her previous best rank, then he awards him/her a lollipop (students love lollipops). Note that the first one in the queue will always get a lollipop and the students arrange themselves in random order in the queue.

What is the expected number of lollipops the headmaster will have to distribute among students if the total number of students in the class is 69? Note that the answer can be a fractional number.

Looking for a little extra cash?

Solve this month's CodeChef Challenge puzzle and send in your answer to codechef@efyindia.com <<mailto:codechef@efyindia.com>>. Three lucky winners get Rs 1,000 in cash! 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to www.codechef.com for the November Challenge that takes place from November 1 to 11, 2010, to win cash prizes up to Rs 20,000. The Direct! "Go for Gold Project" is an ambitious project designed to help an Indian team win the ACM-ICPC World finals. Find out more at www.codechef.com/goforgold.